

(TR-104) PYTHON WITH AI/ML

Training Day 70 Report

Date: 17 April 2026

Topic Covered: Online Door Configurator Research, 3D Visualization & Agentic Workflow Development

The seventieth day of training focused on researching and analyzing the architecture of an Online Door Configurator system along with exploring 3D visualization technologies and workflow-based AI systems. The training also included workflow debugging, validation, and preparation of a structured AI learning roadmap.

- Researched and analyzed the concept and workflow of an Online Door Configurator system.
- Studied the Ryterna Entry configurator and explored its customization workflow and features.
- Analyzed advanced door customization capabilities provided by the Inotherm configurator.
- Identified important system features including live preview, facade upload, inquiry handling, and real-time customization.
- Documented customization options such as door models, RAL colour selection, handles, glass types, and sizing configurations.
- Defined frontend, backend, database, and hosting requirements for the configurator system.
- Analyzed security requirements including secure file handling, CAPTCHA integration, and data encryption.
- Outlined performance requirements for real-time rendering and fast-loading preview systems.
- Researched Three.js for implementing interactive 3D door previews in web applications.
- Explored React Three Fiber for integrating Three.js-based 3D rendering in React projects.
- Understood how 3D models are used for browser-based real-time customization systems.

- Prepared a structured 45-day AI learning schedule divided into multiple modules.
- Included foundational AI concepts, Python programming, Machine Learning, Deep Learning, NLP, Generative AI, RAG systems, and workflow automation topics.
- Designed hands-on projects and implementation-based learning activities for each module.
- Created and configured multiple AI workflows for content generation use cases.
- Integrated workflow components including Input, Prompt Template, Agent, and Output nodes.
- Tested workflow execution flow and validated component connectivity.
- Identified and analyzed workflow execution issues during testing.
- Verified agent configuration, prompt linking, and model setup correctness.
- Reviewed poster content for spelling, grammar, and overall presentation quality.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/workflows>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/designing-an-intelligent-content-generation-workflow-ce1b91fd313b>

Overall, the training strengthened understanding of configurable web systems, 3D visualization technologies, AI workflow systems, workflow debugging, and structured AI learning roadmap preparation.